

A hand is pointing at a tablet screen. The screen displays a bar chart with several vertical bars of varying heights. The bars are rendered as wireframes, showing their 3D structure. The background is a blurred image of a person in a blue shirt.

Credit Card Customer Analytics & Strategic Growth Model

Advanced Data Processing, Segmentation & Predictive Scenarios

Burak Canlı | 2026

Total Customers

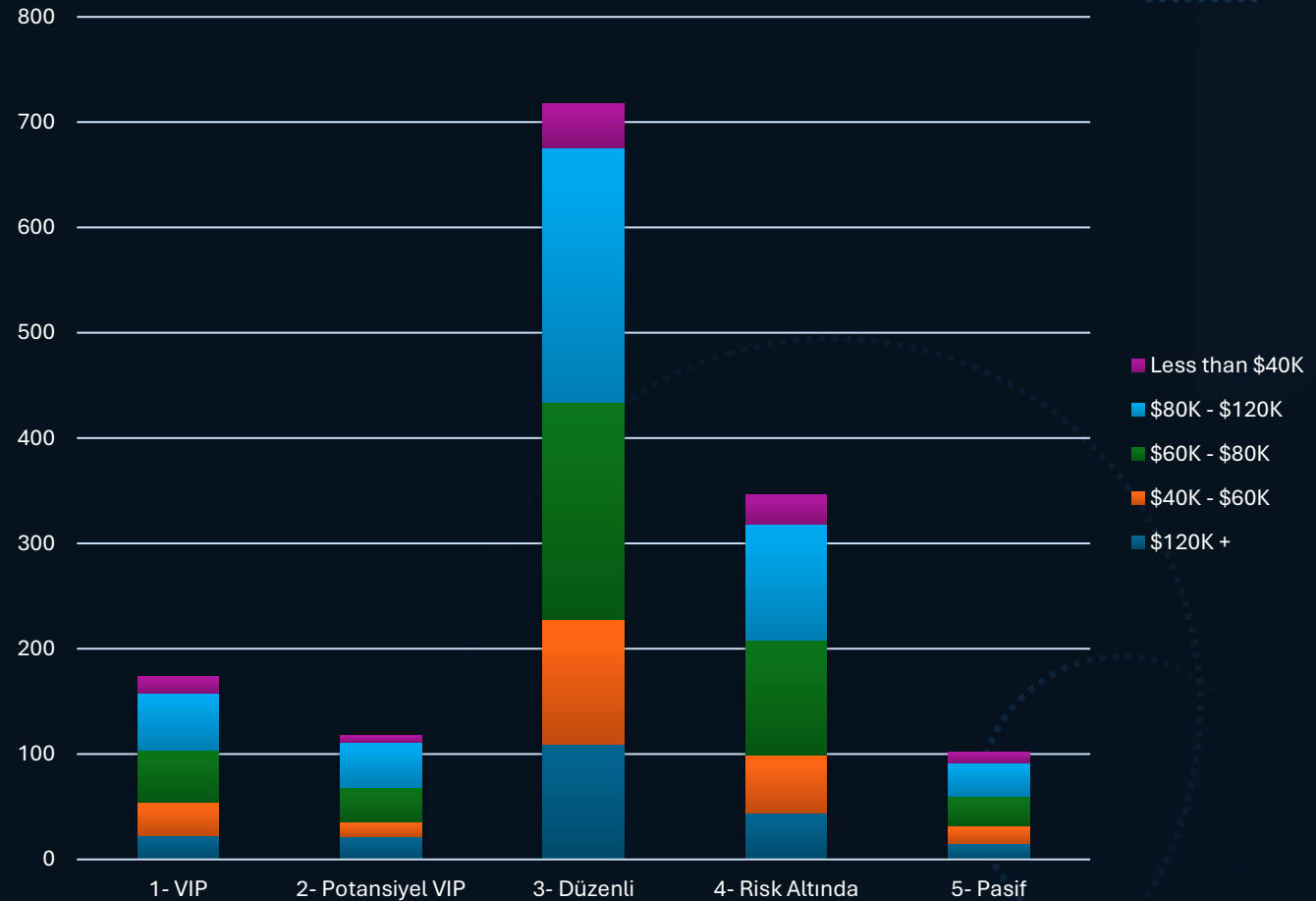
1.458

Total Transaction Volume

\$6.653.199

- Comprehensive cleaning and algorithmic imputation of 10,000+ customer records.
- Implementation of 5-Tier Strategic Customer Segmentation (VIP, Potential, Regular, At-Risk, Passive).
- Data-driven financial simulation and automated C-Level dashboard architecture.

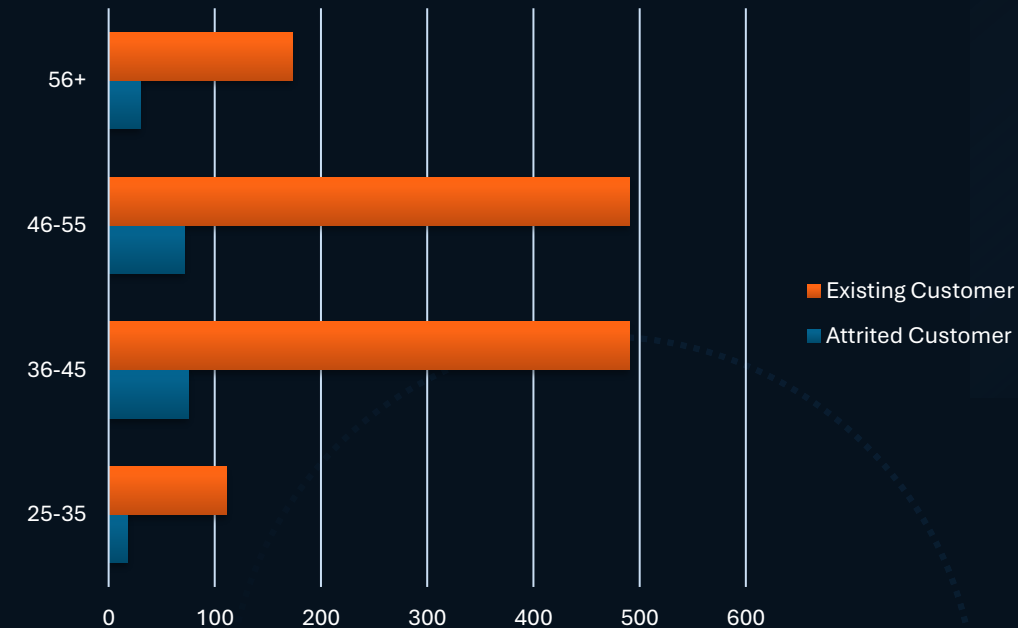
- Algorithmic Hierarchy: Developed a multi-variable model combining transaction volume and frequency rather than simple spending.
- VIP Identification: Successfully isolated high-value cardholders to prioritize retention campaigns.
- Risk Mitigation: Grouped declining and inactive customers into targeted recovery brackets.



5-Tier Strategic Customer Segmentation

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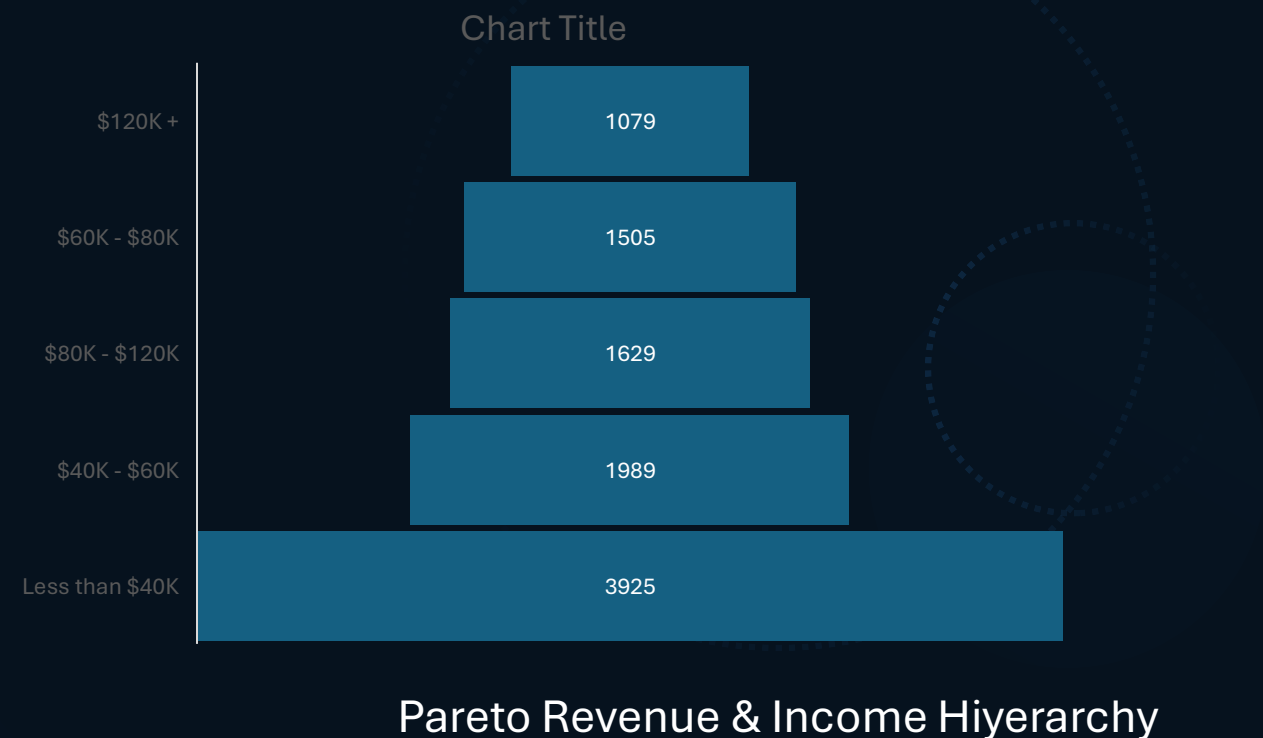
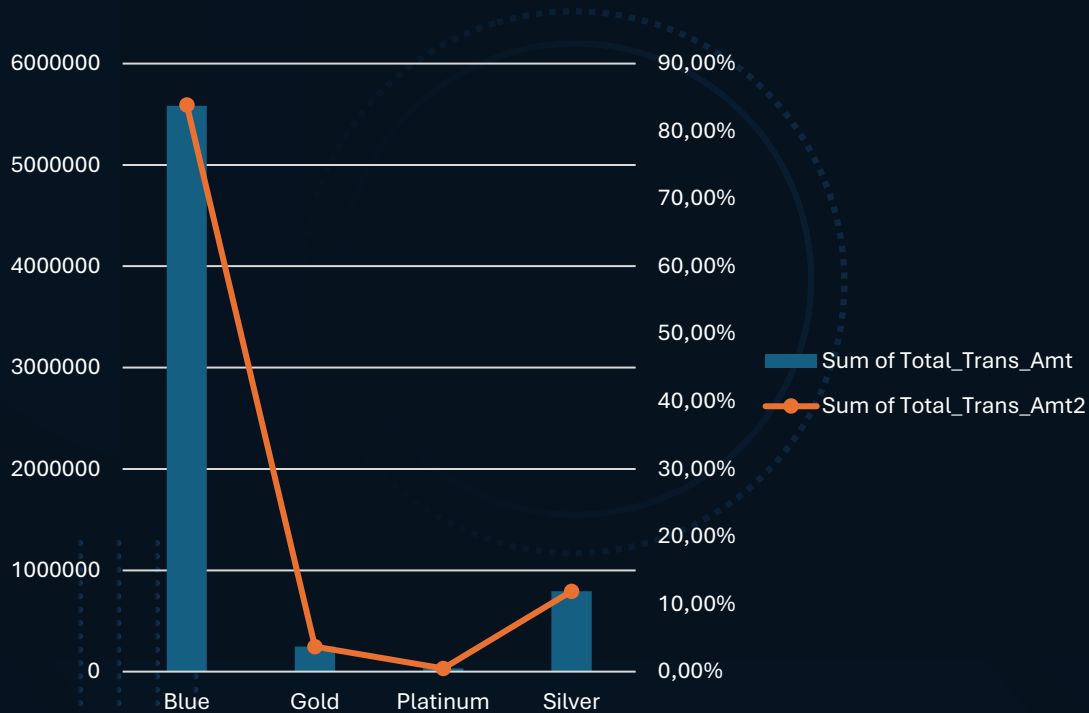
- Age Distribution: Mapped customer density across age brackets (Age_Group).
- Tenure Impact: Evaluated customer loyalty duration against churn rates
- Actionable Insights: Discovered critical drop-off points in specific age tiers.



Count of CLIENTNUM		Column Labels		
Row Labels		F	M	Grand Total
25-35			466	436
36-45			1977	1765
46-55			2208	1952
56+			707	616
Grand Total			5358	4769

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- Core Driver Identification: The top 20% of card categories generate the vast majority of volume (Left Chart).
- Funnel Analysis: Visualized the income bracket hierarchy to understand demographic purchasing power (Right Chart).



Monthly Transaction Volume	
Average Credit Limit	11111,11111
Credit Utilization Rate	0,27
Expected Monthly Transaction Volume	3000

- Targeting: Simulated required average credit limits to achieve a baseline \$3,000 monthly transaction volume.
- Mathematical Baseline: Identified \$11,111 as the optimal credit limit threshold for target acquisition.
- Strategic Constraint: Modeled dynamic limit adjustments for rapid executive decision-making.

	Potential Credit Limits				
3000	5000	8500	10000	15000	
10%	\$ 500	\$ 850	\$ 1.000	\$ 1.500	
20%	\$ 1.000	\$ 1.700	\$ 2.000	\$ 3.000	
30%	\$ 1.500	\$ 2.550	\$ 3.000	\$ 4.500	
40%	\$ 2.000	\$ 3.400	\$ 4.000	\$ 6.000	

- 2D Matrix Architecture: Cross-tabulated prospective credit limits against diverse utilization rates using Data Tables.
- Dynamic Forecasting: Generated multiple potential revenue outcomes instantaneously without altering raw data.
- Risk/Reward Balance: Provided a clear strategic roadmap for limit enhancement campaigns.

- One-Click Refresh: Implemented VBA macro scripting for automated end-to-end data pipeline updates.
- Dynamic Interactivity: Integrated Slicers and offset parameters for real-time dashboard filtering.
- Form Controls: Engineered an interactive risk-premium toggle using Checkbox and dynamic arrays.

Gender

F

M

Refresh Data

☒ Check	TRUE
550	

Education_Level_Clean

College

Doctorate

Graduate

High School

Post-Graduate

Uneducated

Unknown



- Pareto analysis shows a massive dependency on the Top 20% cardholder tier.
- Launch a "Premium Concierge" program with zero-fee transactions for high-volume users.
- Securing the core revenue stream and increasing lifetime value (LTV) by 15%.

Strategy 1: VIP Retention & Loyalty Program

- High attrition rate detected in the 36-45 age group (mid-tenure).
- Deploy automated "We Miss You" email sequences and temporary cashback incentives for inactive segments.
- Reduce the monthly churn rate by 4% within the first quarter.



Strategy 2: Proactive Churn Prevention

- Underutilized credit limits in the "Potential" segment are bottlenecking transaction volume.
- Gradually increase the average credit limit to \$11,111 for targeted low-risk profiles, as simulated in Goal Seek.
- Achieve the \$3,000 baseline transaction volume goal and boost interest revenue.



Strategy 3: Credit Limit Optimization